Geervan K

Software Developer, Full Stack & Backend Systems
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Education

Manipal University Jaipur — Jaipur, India
B.Tech in Computer Science and Engineering GPA: 9.34/10 — *2024 – 2028*

Skills

Languages: TypeScript, JavaScript, SQL, Python, C/C++

Backend & Databases: Node.js (Express), Prisma ORM, PostgreSQL, RESTful API Design, JWT Auth

Frontend & State Management: React, Redux Toolkit, Tailwind CSS, Material UI (MUI), Framer Motion

Core Concepts: Relational Database Design, Role-Based Access Control (RBAC), Complex Allotment Algorithms, Data Structures & Algorithms

Tools: Git, GitHub, Vite, Postman

Experience

Software Development Intern — Mar 2025 – Present
Software Development Cell, Manipal University Jaipur — *Jaipur, India*

- Contributing to a production-grade system serving 2000+ students.
- Worked on backend logic including access control and allocation workflows.
- Improved reliability and scalability under concurrent usage with 0 issues in production.

Technical Team Member — Feb 2025 – Mar 2026
CampusAdda Pvt Ltd — *Jaipur, India*

- Coordinated with partner outlets to gather requirements and resolve issues.

Open Source Contributions

Astryx (Meta Open Source ecosystem) [GitHub] [My Commits]

- Authored and merged **19 pull requests** to @astryxdesign/core and @astryxdesign/cli packages of an open-source agentic design system.
- Resolved issues across **React, TypeScript, theming, design tokens, mobile interactions and cross-platform CLI compatibility**, contributing through root-cause analysis, testing, and code review an upper limit of 5 pr's are under review out of which 3 are ready to merge at this moment

Fluent UI (Microsoft Open Source) [GitHub] [My PR]

- Fixed a rendering bug in **@fluentui/react-nav** where the selection indicator on a selected **NavSubItem** inside an expanded **NavCategory** was invisible; **approved by maintainers** and pending merge.
- Diagnosed the root cause as **CSS clipping and indicator offset misalignment**, correcting offsets for standard and small density and removing `overflow: hidden` from **NavSubItemGroup**; verified across Storybook stories, unit tests, lint, and type-checking.

Projects

Elective Management System [Demo] [Private]

- Developed a production-grade,academic system utilizing a modern full-stack architecture (**React, TypeScript, Node.js**) to streamline elective allocation for **2000+ active students**.
- Engineered granular **Role-Based Access Control (RBAC)** across multiple tiers (Student, Professor, Admins) and designed transaction-isolated database allocation algorithms to prevent preference conflicts.
- Optimized database reliability under concurrent high load by implementing indexed **Prisma & PostgreSQL** query patterns, achieving zero data anomalies during peak enrollment windows.

MUJ SLCM App (Unofficial) [Demo] [Private]

- Built and scaled a Flutter app and PWA for MUJ students to access the official SLCM, reaching **4500+** cumulative users to streamline access to academic workflows.
- Reduced login and access time from **20s to 5-7s through authentication and performance optimizations.**
- Improved user experience with persistent sessions, secure credential storage, simplified navigation, and convenient access to attendance and marks.

Antigravity Portable [GitHub]

- Architected a local-first portable development environment (**PWA + Node.js**) accessible from any device through **Cloudflare Tunnels**, eliminating the need for dedicated cloud workspaces.
- Developed a custom HTTP/WebSocket reverse-proxy with real PTY terminal streaming, securely routing local development servers (Next.js, Vite, FastAPI, Flask) over a single authenticated endpoint.
- Built real-time diagnostics featuring multi-language error parsing, automatic project stack detection, and secure workspace sandboxing with Bearer authentication and path isolation.

Synapse: Zero-Knowledge IDE Memory Bridge [Marketplace] [GitHub]

- Built and published a local-first, zero-knowledge sync tool as a **VS Code Extension** and **Chrome MV3 Extension**, connecting browser LLMs directly to the local IDE.
- Implemented **AES-256** local encryption (Crypto-JS) to ensure code snippets, schemas, and design documents remain private on the developer's machine.
- Integrated **Transformers.js**, **Ollama**, **Supabase (PostgreSQL)**, and **OAuth 2.0** to enable offline AI processing, real-time synchronization, and secure authentication.

Patent

Adaptive Learning from Presentation Slides via Multi-Modal Content Generation and Concept Graph Modeling Published Sep 2026

* Co-inventor on **Indian Patent Application No. 202611099581 A** (filed 18/08/2026, published in Patent Office Journal No. 39/2026), with applicant **Manipal University Jaipur**.
* Designed a system that converts presentation slides into personalized learning using **concept dependency graphs**, **adaptive learning-mode scheduling**, and **multi-modal content generation** (summaries, assessments, visualizations, explanations).
* Included **content-hash-based deterministic caching** and a **feedback-driven generative validation loop** to refine outputs and update learner profiles and knowledge gaps.